

Program : Diploma in Communication and Computer Networking	
Course Code : 6329B	Course Title: Digital Communication lab
Semester : 6	Credits: No Credit
Course Category: Engineering Science	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- To explore digital communications on bread board and to simulate it with **Python/MATLAB/ SCILAB/ LabVIEW**
- To experiment on analog to digital conversion, modulation and pulse shaping

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Fundamentals of Electrical and Electronics Engineering	2039	Fundamentals of Electrical and Electronics Engineering Lab	6

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Setup simple prototype circuits for waveform coding and digital modulation on breadboard or trainer kits.	12	Applying
CO2	Simulate the digital communication experiment using Python/MATLAB/ SCILAB/ LabVIEW	15	Applying
CO3	Demonstrate the skills to use modern engineering tools, software's and equipment to analyze problem	12	Applying
	Lab Test	6	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3						
CO3	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Setup simple prototype circuits for waveform coding and digital modulation on breadboard or trainer kits.		
M1.01	To study signal sampling and its reconstruction	6	Applying
M1.02	To construct and study a BASK generating circuit	6	Applying
CO2	Simulate the digital communication experiment using Python/MATLAB/ SCILAB/ LabVIEW		
M2.01	To construct and study the characteristics of BPSK modulation and demodulation	3	Applying
M2.02	To construct and study a PCM transmitter and receiver	3	Applying
M2.03	To study the characteristics of delta modulation and demodulation	3	Applying
M2.04	To study the Time Division Multiplexing (TDM) and draw its waveforms.	3	Applying
M2.05	To simulate Quadrature Phase shift keying technique using MATLAB software	3	Applying
	Lab Test – I	3	
CO3	Demonstrate the skills to use modern engineering tools, software's and equipment to analyze problem		

M3.01	Familiarization with Software Defined Radio (Hardware and Control Software) for reception and transmission of RF signal.	6	Applying
M3.02	Familiarize how it can be interfaced with computer	3	Applying
M3.03	Familiarize with GNU Radio (or similar software's like Simulink/ LabView) that can be used to process the signals received through the SDR hardware.	3	Applying
	Lab Test – II	3	

Text / Reference

T/R	Book Title/Author
T1	Mathuranathan Viswanathan, "Digital Modulations using Python"
T2	WH Tranter, KS Shanmugan, TS Rappaport, KL Kosbar, "Principles of Communication Systems Simulation with Wireless Applications", Prentice Hall
R3	Carl Laufer, "The Hobbyist's Guide to the RTL-SDR: Really Cheap Software Defined Radio"

Online Resources

Sl.No	Website Link
1	Neel Pandeya, "Implementation of a Simple FM Receiver in GNU Radio," https://kb.ettus.com/
2	Michael Ossmann, "Software Defined Radio with HackRF," YouTube Tutorial